

Abhay Tomar

Delhi | GMail | +91 9582978227 | Portfolio | LinkedIn | GitHub

Summary

As an Embedded Software Engineer with experience in developing firmware and embedded applications for microcontrollers like ESP32, ARM and TI. Skilled in C/C++, real-time operating systems (RTOS), and hardware-software integration. Strong background in communication protocols (UART, SPI, I2C, CAN) and peripheral interfacing. Delivered innovative projects like an Stepper Motor Controller using ESP32 and a Magnetic Meter using MSPMOL1306. Passionate about optimizing system performance and developing reliable embedded solutions for industries such as IoT and automotive. Dedicated to delivering efficient and scalable solutions in embedded systems. Currently exploring opportunities in automotive embedded systems.

Experience

Embedded Software Engineer, SVURG Digital System Solutions – Jasola, Delhi Sep 2024 – Present

As an Embedded Software Engineer at SVURG Digital System Solutions, developed and optimized software for ESP32, ESP8266, MSP430, and MSPMOL1306 microcontrollers. Responsibilities included implementing real-time firmware in C/C++, integrating peripherals, debugging with JTAG, and maintaining documentation, while employing best practices in version control.

Education

IGNOU University, Bachelor in Computer Applications July 2020 – June 2024

- GPA: 7/10
- **Coursework:** Computer Architecture, Comparison of Learning Algorithms, Computational Theory

Certifications

Git/GitHub: By Great Learning (Link) July 2024

Technologies

Languages: C++, C, Embedded C

Protocols: SPI, UART, I2C, CAN

IDE: Arduino, CCS Theia, STM32CubeIDE

Projects

Stepper Motor Controller (ESP32)

- Developed an embedded system to control a stepper motor using ESP32
- Implemented a TFT touchscreen UI for motor control with real-time feedback.
- Designed hierarchical menu navigation using a rotary encoder and button controls.

Magnetic Meter (AS5600 and MSPMOL1306)

- Developed an Embedded system to use this as a feedback system for CNC Machine for checking X, Y and Z coordinates using AS5600 magnetic sensor and MSPMOL1306 controller.
- **Project Objective:** Use as feedback system for a high-performance CNC controller using the STM32H747 microcontroller, leveraging its dual-core architecture for real-time multitasking. The M7 core handles motion control, while the M4 core manages peripheral communication, ensuring precision, efficiency, and scalability while improving system responsiveness and integration with CNC hardware and software.